

ENGINEERING REBOOT 2026

Week 03 – Day 12 Field Manual

Collection Traversal & Iteration

Version 1.3 – Re■Baselined Edition – Publication Edition

Status: LOCKED

PHASE 0 – MISSION BRIEF

Objective

Learn how software systems process collections using iteration.

Day 11 introduced storage of collections.

Day 12 introduces traversal of collections.

This is one of the most important concepts in software development because nearly every real■world s

Core Concepts

- for loops
- traversal
- iteration
- counters
- collection processing

Engineering Applications

Processing User Accounts

```
users = ["alice", "bob", "charlie"]
```

Processing Sensor Readings

```
temperatures = [71, 73, 75, 72, 70]
```

Processing Server Inventory

```
servers = ["web01", "web02", "db01"]
```

Learning Objectives

- Traverse a list using a for loop
- Process each item in a collection
- Count collection elements
- Generate reports from collections
- Apply collection traversal to real-world datasets

Deliverable Inventory

1. iterate_names.py
2. number_list.py
3. grade_report.py
4. Notes.md
5. Journal.md
6. Evidence Package

PHASE 1 – ITERATE NAMES

Filename

iterate_names.py

Implement

Create:

```
names = ["Alice", "Bob", "Charlie", "David"]
```

Use a for loop to display each name on its own line.

Validation Matrix

Expected Output

Alice

Bob

Charlie

David

Evidence Checkpoint

iterate_names-Code.png

iterate_names-Output.png

Completion Checkpoint

■ Exercise Completed

■ Validation Completed

■ Evidence Captured

PHASE 2 – NUMBER LIST

Filename

number_list.py

Implement

Create:

```
numbers = [10, 20, 30, 40, 50]
```

Use a for loop to:

- Display each number
- Calculate a running total

Display:

- All numbers
- Final total

Validation Matrix

Expected Output

10

20

30

40

50

Total: 150

Evidence Checkpoint

number_list-Code.png

number_list-Output.png

Completion Checkpoint

■ Exercise Completed

■ Validation Completed

■ Evidence Captured

PHASE 3 – GRADE REPORT

Filename

grade_report.py

Implement

Create:

```
grades = [88, 92, 79, 95, 84]
```

Use a for loop to:

- Display each grade
- Count grades
- Calculate average

Display:

Grades: 88, 92, 79, 95, 84

Count: 5

Average: 87.6

Validation Matrix

Expected Output

Grades: 88, 92, 79, 95, 84

Count: 5

Average: 87.6

Evidence Checkpoint

grade_report-Code.png

grade_report-Output.png

Completion Checkpoint

■ Exercise Completed

■ Validation Completed

■ Evidence Captured

PHASE 4 – NOTES

Create

03-Week-03

■■■ Day-12

■■■ Notes

■■■ Notes.md

Required Sections

- Concepts Learned
- Key Commands
- Key Observations

- Lessons Learned
- Questions For Future Study

Suggested Topics

- What is traversal?
- How does a for loop work?
- Why are loops important?
- How are loops used in software systems?

Template

Concepts Learned

Key Commands

Key Observations

Lessons Learned

Questions For Future Study

Completion Checkpoint

■ Notes Completed

PHASE 5 – JOURNAL

Create

03-Week-03

■■■ Day-12

■■■ Journal

■■■ Journal.md

Required Sections

- Summary
- Challenges Encountered
- Solutions Applied
- Confidence Assessment
- Reflection
- Tomorrow Preparation

Suggested Reflection

Why is processing a collection more powerful than processing individual variables?

Template

Summary

Challenges Encountered

Solutions Applied

Confidence Assessment

Reflection

Tomorrow Preparation

Completion Checkpoint

■ Journal Completed

PHASE 6 – GIT OPERATIONS

Execute

git status

git add .

git commit -m "Complete Day 12 iteration exercises"

git push

git fetch

git status

git branch -vv

Preferred End State

Your branch is up to date with 'origin/main'

nothing to commit, working tree clean

Evidence Checkpoint

Git-Operations.png

Must Show

git add .
git commit
git push
git fetch
git status
git branch -vv

Completion Checkpoint

- Git Operations Completed
- Repository Synchronized
- Evidence Captured

END-OF-DAY SUCCESS CRITERIA

- iterate_names.py completed
- number_list.py completed
- grade_report.py completed
- All validation completed
- Exercise evidence captured
- Notes completed
- Journal completed
- Git workflow completed
- Git-Operations.png captured
- Remote synchronization verified
- Working tree clean

DAY 13 PREVIEW

Topic

Dictionaries & Structured Data

Expected Deliverables

- dictionary_basics.py
- student_record.py
- inventory_dictionary.py

Core Concepts

- dictionaries
- keys
- values
- records
- structured data

Engineering Focus

Representing real-world entities using key-value data structures.